3. Architecture Document (Expanded)

Introduction:

The **Architecture Document** provides a technical breakdown of how the Power BI dashboard is designed and implemented. It includes the framework of the project, how Power BI components interact with each other, and how data flows from external sources into the Power BI system for analysis.

The document should cover aspects such as the **Power BI Architecture**, **Data Integration**, **Power BI Service**, and **Data Sources**. It aims to explain the system's design at a granular level, ensuring that everyone working on the project understands how components fit together.

Scope:

The scope of this architecture document is to define the overall technical infrastructure of the Power BI dashboard. This includes:

- **Power BI Architecture**: The workflow from importing data to building reports.
- **Data Integration**: Describing how the data is fetched, transformed, and loaded into Power BI.
- **Visualization Framework**: Detailing how the dashboard is structured for reporting, with an emphasis on how users interact with the reports.
- **Data Sources**: Identifying the databases, cloud storage, or other data repositories used to fetch the Amazon sales data.

Power BI Architecture:

The **Power BI Architecture** can be broken down into key stages:

1. **Data Ingestion**:
    - Raw sales data is ingested from various sources, such as SQL databases or CSV files, using Power BI Desktop's connectors.
    - Data is brought into Power BI from external sources and placed into tables that will be used for analysis.
2. **Data Transformation (Power Query)**:
    - Power Query is used to clean, shape, and transform the data. This could involve parsing dates, handling missing values, normalizing column formats, and merging data from different sources into a single, unified dataset.
    - The data model is created by defining relationships between tables.
3. **Data Modeling (Relationships and DAX)**:
    - Power BI uses **relationships** between tables to link data sources together (e.g., linking sales data to product categories, regions, etc.).

  - **DAX (Data Analysis Expressions)** is used to create calculated fields and measures, such as total revenue, total profit, and monthly trends.

4. **Data Visualization**:
   - Visualizations like bar charts, line charts, pie charts, and tables are created to present the key performance indicators (KPIs) clearly. Slicers are added to enable users to filter data by time, region, and other variables.

5. **Power BI Service**:
   - The report is published to the **Power BI Service**, which allows the data to be shared across different stakeholders in an organization. Users can view and interact with the reports in a web-based environment.
   - **Scheduled Refresh**: Data is refreshed automatically in Power BI Service based on predefined schedules.

Power BI Service Architecture:

The **Power BI Service** allows for seamless sharing, collaboration, and accessibility of reports and dashboards.

- **Workspaces**: Reports are stored and shared in specific workspaces, ensuring the appropriate permissions are applied.
- **Dashboards**: Multiple reports can be combined into a single, interactive dashboard, offering a holistic view of the data.
- **Real-Time Collaboration**: Users can collaborate by sharing dashboards and interacting with them in real-time.
- **Mobile Access**: Reports and dashboards are accessible through mobile apps, providing decision-makers the flexibility to access data on the go.

Data Server:

The **Data Server** is where the raw sales data is stored. This could be:

- **Cloud Database**: Data hosted on cloud services such as AWS, Azure, or Google Cloud.
- **On-Premises Database**: Data stored locally on a company server, usually accessed through a **Power BI Gateway** to facilitate secure data refreshes.

Power BI fetches data from these sources, processes it through transformations, and then makes it available for reporting and visualization.